

Exercise Sheet 1

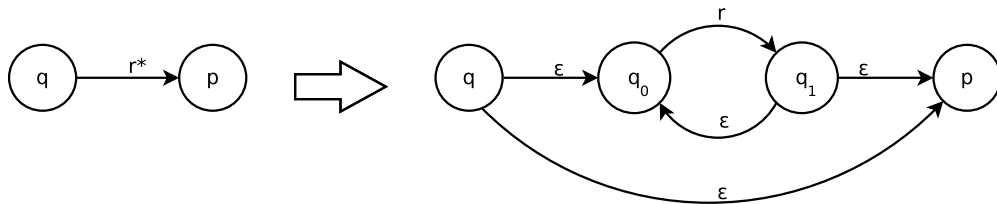
Exercise 1.1 Regular Expressions and Languages

Give a regular expression that *fully* describes *floating constants* in C99. Note that you also have to cover more esoteric cases such as hexfloats!

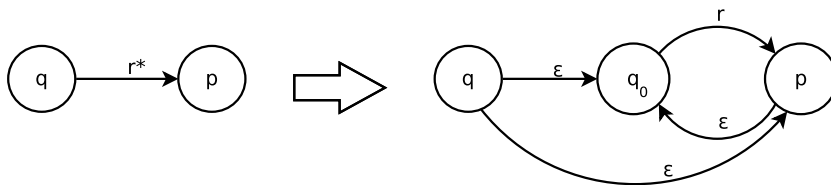
See <http://www.open-std.org/JTC1/SC22/WG14/www/docs/n1256.pdf>, §6.4.4.2.

Exercise 1.2 From Regular Expressions to Finite State Machines

In the lecture, the following rule was given for constructing a NFSM from the regular expression r^* :



Provide a counterexample showing that this rule would be incorrect:



Exercise 1.3 Regular Expressions and Languages

The lecture defined regular expressions using the metacharacters $\emptyset$ and ε . Show that they are the neutral elements with respect to the alternative and concatenation operations in regular expressions. This means show that:

- $(r_1 | \emptyset)$ describes the same language as r_1
- $(r_1 \varepsilon)$ describes the same language as r_1

only by reasoning about the described languages as shown in the lecture. Assume the regular expression r_1 to denote the language R_1 .

Exercise 1.4 Language Operations

Let L be an arbitrary language. Which element(s) are contained in $X = L^* - L^+$?

If X is identical for any L , please argue why. Otherwise describe X depending on L .